

Implementing electronic commerce in SMEs — three case studies

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In recent times electronic commerce has been accepted as a bona fide business practice in the commercial world. However, most research and news coverage has focused upon the usage and practices of large firms. In 1996 the Electronic Commerce Innovation Centre, at Cardiff University, secured funding from the European Regional Development Fund and BT to manage a two-year project aimed at researching electronic commerce implementations in small-to-medium-sized enterprises. In late 1998 funding was received from BT to extend the project for a further two years.

The aim of the project is twofold — firstly, to undertake the implementation process of widely available electronic commerce solutions, and, secondly, to monitor and research the usage patterns and the effect of these technologies not only on the smaller companies independently, but also on their customers, suppliers and collaborators (i.e. the respective supply chains). Resulting from this research, the paper provides three case studies of the electronic trading activities of smaller firms based in South Wales.

1. Introduction

This paper addresses two key components of the supply chain — electronic commerce (eCommerce) and the role of small-to-medium-sized enterprises (SMEs). The paper shows how SMEs are using eCommerce in their everyday business practices in order to integrate more fully into their respective supply chains. These SMEs not only offer examples of small businesses operating at a local or indeed regional level, but also with national and international supply chains. The SMEs discussed are all members of a test platform project currently being conducted in South Wales by the eCIC, at Cardiff University.

Three specific case studies are provided which highlight how eCommerce, and specifically the Internet, have become central to a number of activities in each of the SMEs. The benefits and issues of eCommerce for SMEs, in respect of supply chain management, are examined. From this discussion the real requirements for SMEs to obtain a more integrated position with both customers and suppliers are revealed.

2. The South Wales eCommerce test platform project

In 1996 the European Regional Development Fund (ERDF) awarded funding to eCIC to manage a test platform project to examine the effect of implementing eCommerce in SMEs. BT and Cardiff University matched this funding. Fifteen SMEs based in South Wales were chosen for involvement. The project has also gained significant local government support in South Wales and in so doing

established a unique test platform for eCommerce technologies involving private and public organisations. This was exemplified in late 1998 when BT agreed to fund eCIC for a further two years in order to continue this project (albeit with a reduced number of SMEs, ten of the original fifteen are now supported).

The project, 'Electronic Commerce in Support of Domestic and International Trade', is providing a test platform for determining various ways eCommerce can support different trading scenarios across the supply chains in which SMEs operate. In particular, it has already demonstrated and evaluated a wide range of eCommerce technologies on user sites. Supporting activities are investigating and researching the most suitable ways in which businesses can apply eCommerce technologies, migrate from paper-based systems and identify the benefits eCommerce can bring to SMEs.

The fifteen SMEs involved in the project form a diverse group, representative of the South Wales and wider UK industrial base. Thirteen of the companies can be classed as manufacturers with one in the service sector and one in transport. The companies cover the broad geographical region of industrial South Wales (ISW) (see Fig 1), formerly the coal-mining heartland, and range in size from four to two-hundred employees. Seven of the SME manufacturers are regular exporters, with exports as a percentage of turnover ranging from 10% to over 95%; one company operates in the international transport arena, three operate in the UK-wide market and four in local markets.

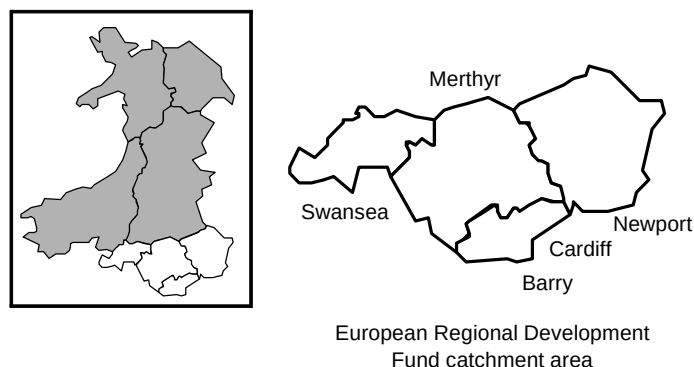


Fig 1 The project region

3. Case studies of electronic commerce and supply chain management

3.1 *Electronic commerce in the supply chain*

The term 'electronic commerce' relates to a distinct subset of information and communications technologies (ICT) which specifically addresses the development of technologies used for directly communicating information and data between organisations. The term 'ICT' has only recently emerged from the rapidly merging sectors of information technology (formerly computing and before that data processing) and communications (derived from telecommunications). The term 'information technology' itself has been widely recognised as a strategic weapon for SMEs, especially in maintaining a competitive advantage [1, 2].

These technological developments include commercial transactions not only from commercial business-to-business interactions but also from business-to-government and from non-profit organisations. The role of the government is vital in both promoting eCommerce, and also in proactively participating in electronic trading, as noted by MacLeod [3]. The following definition is given to provide a clear appreciation of eCommerce:

'Electronic commerce is the transacting of business information and data, between businesses and between customers, in order to more effectively and efficiently complete processes across the spectrum of a business.'

It can be seen from Fig 2 that eCommerce has potential impact throughout the supply chain with a host of trading partners [4]. The central box in Fig 2 shows a typical manufacturing company, with traditional internal applications and communications across various departments and functions. However, external communications and interactions are increasingly being conducted via eCommerce applications, such as classical electronic data interchange (EDI) with suppliers on a global basis, and consumer-oriented eCommerce with customers.

As a point of note, classical EDI has been the focus of numerous papers. Typically, EDI has been implemented in so-called hub-spoke relationships where all too often the benefits are more dramatic for the hubs (i.e. usually larger, more dominant customer or supplier), for example, within the realms of port operations [5], retail supply and distribution [6], and engineering supply chains. The three case studies in this paper exemplify a combination of the eCommerce applications outlined above, which offer a broader remit than just EDI.

The following definition of supply chain integration particularly encompasses the role of SMEs, in addition to larger firms, in the supply chain [7]:

'... a generic term that encompasses the co-ordination of order generation, order taking, and order fulfilment/distribution of products, services, or information material suppliers, channel supply partners (wholesalers/ distributors and retailers), and customers themselves are all key players in the supply chain management ...'

However, Stewart [8] argues that '... the traditional supply chain — of product development, procurement, production, sales, distribution and field service — is therefore not a chain at all.' Stewart points out that the results of this functional orientation are to introduce error and distortion, add unnecessary cost and undermine competitive positioning. This is because the functional organisation isolates internal departments, suppliers and customers.

Although the remit of supply chain integration, according to Stewart, is beyond the scope of this paper, it can be seen from Kalakota and Whinton's definition [7] and Fig 2 that eCommerce certainly has a pivotal role to play in the schema of supply chain integration for SMEs. Tan, Kannan and Handfield [9], who describe supply chain management as focusing on how firms utilise their suppliers' processes, technology, and capability to enhance competitive advantage, further compound this. As will be discussed in the following sections, eCommerce is increasing in its importance as part of this technology element.

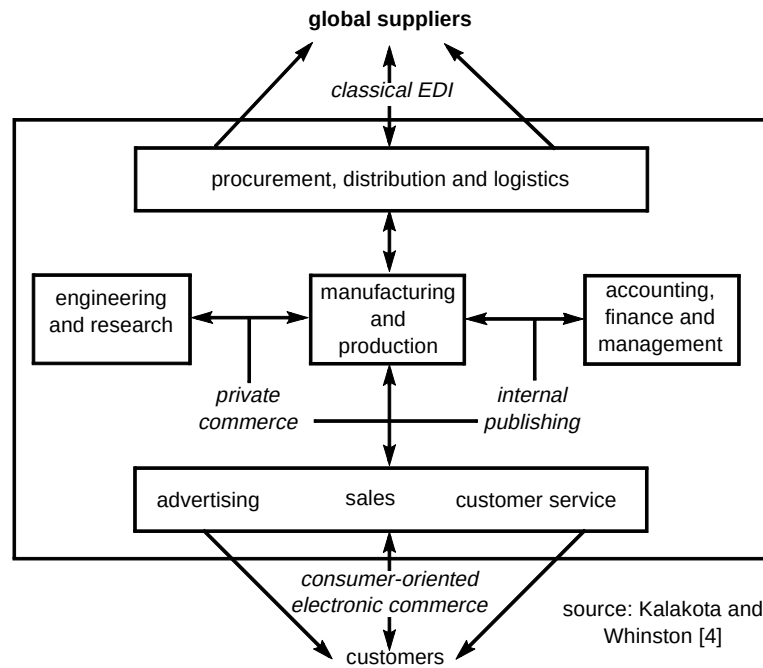


Fig 2 Electronic commerce applications in the supply chain.

3.2 Three case study SMEs

The SMEs were chosen to meet various criteria, including:

- to cover the geographic catchment area of ISW,
- to provide a range of domestic and international traders,
- to represent business-to-business and business-to-consumer traders.

In addition, the SMEs form service providers and product manufacturers, and the criteria ensured that there is at least one cluster of companies within the fifteen which trade together, i.e. intra-business.

Table 1 provides a profile of three of the fifteen SMEs involved in the test platform project. These companies have been chosen for detailed study because they offer three entirely different industrial sectors, they cover the geographic region as previously outlined, and they are deemed 'typical' SMEs in terms of employment size. Although Company A does undertake small manufacturing activity, such as hose and couplings fittings, its core business can be classed as tertiary service activity. Both Companies B and C, however, are manufacturing-based companies.

The two manufacturers are based in single industrial units and Company A operates from a main trading office and depot in the north of ISW, and also has two trading outlets in small towns in the heart of the Valleys region. All three companies are privately owned with annual turnover ranging between £3—4.5 million and with between 50 and

100 employees; they are therefore typically well-established SMEs.

Company A remains a company servicing its immediate locality, although its supplies are sourced from throughout the UK; Company B is a globally operating business, mainly sourcing supplies from UK distributors and selling to numerous export markets; Company C is a UK-wide operator.

Table 2 provides an overview of the eCommerce technologies implemented and tested in each SME. As all three SMEs were technologically capable, and willing to experiment, eCIC decided to supply integrated services digital network (ISDN) lines so that fast data communications and access times to the Internet could be tested. All of the SMEs also had Web sites designed.

Due to the reliance of Company A on catalogues, the company was identified as a potential test case for an e-catalogue development in conjunction with BT — this remains on-going. Company C also showed interest in electronic banking in order to improve their financial management. Both Company B and Company C also have considerable research, design and development processes, which were identified as potentially being enhanced through desktop videoconferencing.

In totality, only five of the fifteen SMEs had e-mail connections at the start of the project, and these were hardly used. Four of the companies had experienced communications with trading partners regarding EDI but none were actually conducting transmissions. In addition, only two of the SMEs had Web sites; therefore the levels of eCommerce experience were minimal.

Table 1 Company profiles of the SMEs.

Function/company	A	B	C
Industry sector	Industrial supplies	Telecommunications test equipment manufacturer	Specialist cable harness manufacturer
Operational locations	Ebbw Vale, plus 2 trading outlets	Newport	Port Talbot
Group holding	Privately owned	Privately owned	Privately owned
Annual turnover	£3.4m	£4.5m	£3m
Number of staff	50	56	85
Customer base	South Wales only, mainly blue-chip	>50% exports, resident in UK, mainly blue-chip	UK wide, mainly blue-chip
Major customers	AIWA, AZCOM, Carpets International, Sony	BT, COMTECH, Mercury, RS Components	Avery Berkel, Control Techniques, Four Square
Supplier base	UK wide, mainly OEMs and large distributors	Global, mainly OEMs	UK wide, mainly OEMs
Major suppliers	Bosch, Spicers, Stanley	Quantum Electronics	AMP, Molex, RS Components
Role in supply chain	Key distributor to local market	Unique position in global market	Key supplier in manufacturing cycle
eCommerce usage to support supply chain	Internet for sourcing and communications	Internet for R&D and customer service via Web	Internet for communications

Table 2 eCommerce technologies implemented in each SME.

Company	Communications type		Other electronic commerce technologies				
	ISDN	Modem (PSTN)	Web site*	EDI**	E-catalogue	E-banking	Desktop videoconferencing
A	✓	—	✓	—	✓	—	—
B	✓	—	✓	✓	—	—	✓
C	✓	—	✓	✓	—	✓	✓

* includes Internet access and e-mail

** includes file exchanges (e.g. CAD/CAM)

3.3 Information flows

Figure 3 shows the major information flows within Company A and with external trading partners. By volume the majority of information flows occur between the sales teams and customers.

Figure 4 shows the major information flows within Company B and with external trading partners. In comparison with Company A, Company B has a complicated set of external information flows. This is primarily due to Company B's research and development (R&D) department, which has many interactions with sub-contractors, suppliers, customers and distributors.

Company A's operations are far more straightforward, in as much as the sourcing of supplies and the same sales teams undertake interactions with customers. Further, the supplier and customer bases of Company A are localised within South Wales and have not expanded a great deal, in pure volume terms, in recent years.

However, Company B's R&D team constantly engage in communications with new suppliers, of information and products, on a global basis. Company B also has an ever-

expanding customer-base, again on an international scale. The very nature of global trade means that more in-depth interactions are required than for Company A in its local environment. For example, Company B has to deal with Customs and Excise, similar foreign organisations, freight forwarders, shippers and agents.

As will be seen in the following section, the use of eCommerce has assisted the R&D team, in particular in more effectively undertaking these communications. For example, the Internet has provided a communications medium that allows relationships to be initiated and developed quickly, as well as providing a rich source of technical data sheets and manuals. Therefore, the R&D team now regularly uses the Web to obtain technical information, data sheets, software upgrades, competitor information and patent searches. The benefit here for the company is that the R&D team believe they can continue working on a specific project without having to stop, and work on something else, while waiting for a data sheet or technical information to be supplied. This is invariably via an advert seen in a magazine or journal and usually posted from the US, with lead-times averaging around 2 weeks for such documentation.

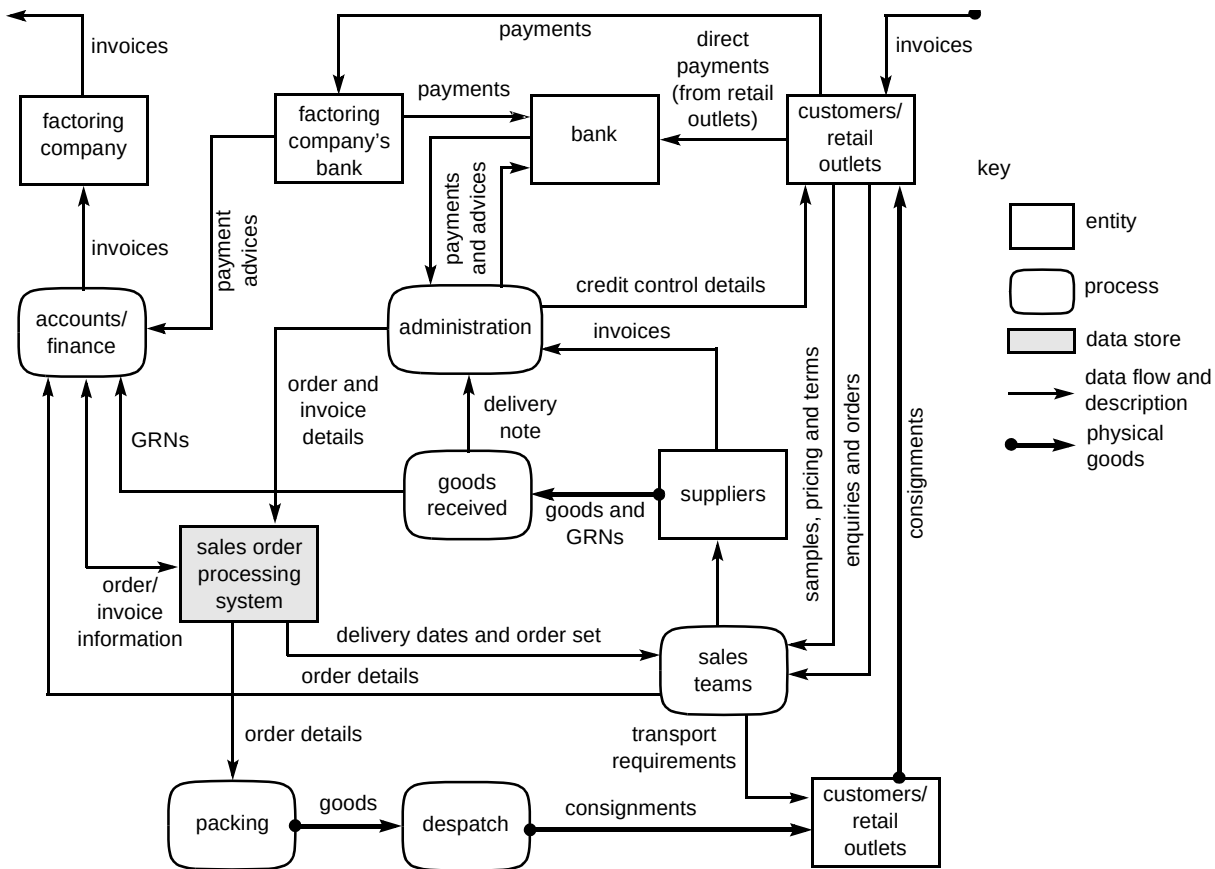


Fig 3 Major inter- and intra-company information flows at Company A.

Therefore the R&D team feels more productive and ultimately can decrease development times for circuit boards, software, hardware (casings) and the like. The mass of information on the Internet, including competitors' products of course, also means that design criteria can be more widely influenced so that products may be built using, for example, new materials and with different techniques.

Figure 5 shows the major information flows within Company C and with external trading partners. Company C's trading environment lies in between those of the former two companies. For example, there are some levels of interactions at the design and development stage, but not as advanced as Company B's. Conversely, there is close liaison with the customer-base which numbers just twelve, far less than that of Company A.

Company C also supplies on a UK-wide basis which invariably means that the close relationships developed on a local basis by Company A cannot be replicated. The supply base of Company C is in the main UK-wide but does also extend overseas, to the US and Europe. As such, many of the interactions as described for Company B are also relevant.

Company C's design and development part of the business (essentially a combination of the Managing and Sales Directors), although not a separate unit as in Company B, requires close interactions with both sides of their supply

chain. The Internet has greatly facilitated this requirement by offering a cheap and easy-to-use communications tool, as well as an information source to aid design work.

The three SMEs thus all have different roles within their supply chains, in terms of the interactions and information flows. Company A can be deemed the most straightforward, a classic localised business operating in a well-established environment where personal interactions are prevalent.

Company B is the most complex due to its product range, which tends to have a long life cycle and thus customers from overseas, in particular, order on the company irregularly. The R&D emphasis of the company also means that communications on a global basis are fundamental to developing new products and servicing the requirements of customers.

Finally, Company C exudes some of the characteristics of the two others. This SME sources supplies locally where possible, but sells and thinks very much on an international basis. It is this element of thought, or attitude, towards commerce that is key in determining whether an SME not only implements, but also benefits, from eCommerce.

4. Benefits of eCommerce for SMEs

The benefits of eCommerce to all fifteen SMEs in their respective supply chains have been fundamental. Many of the companies have found a strong requirement for

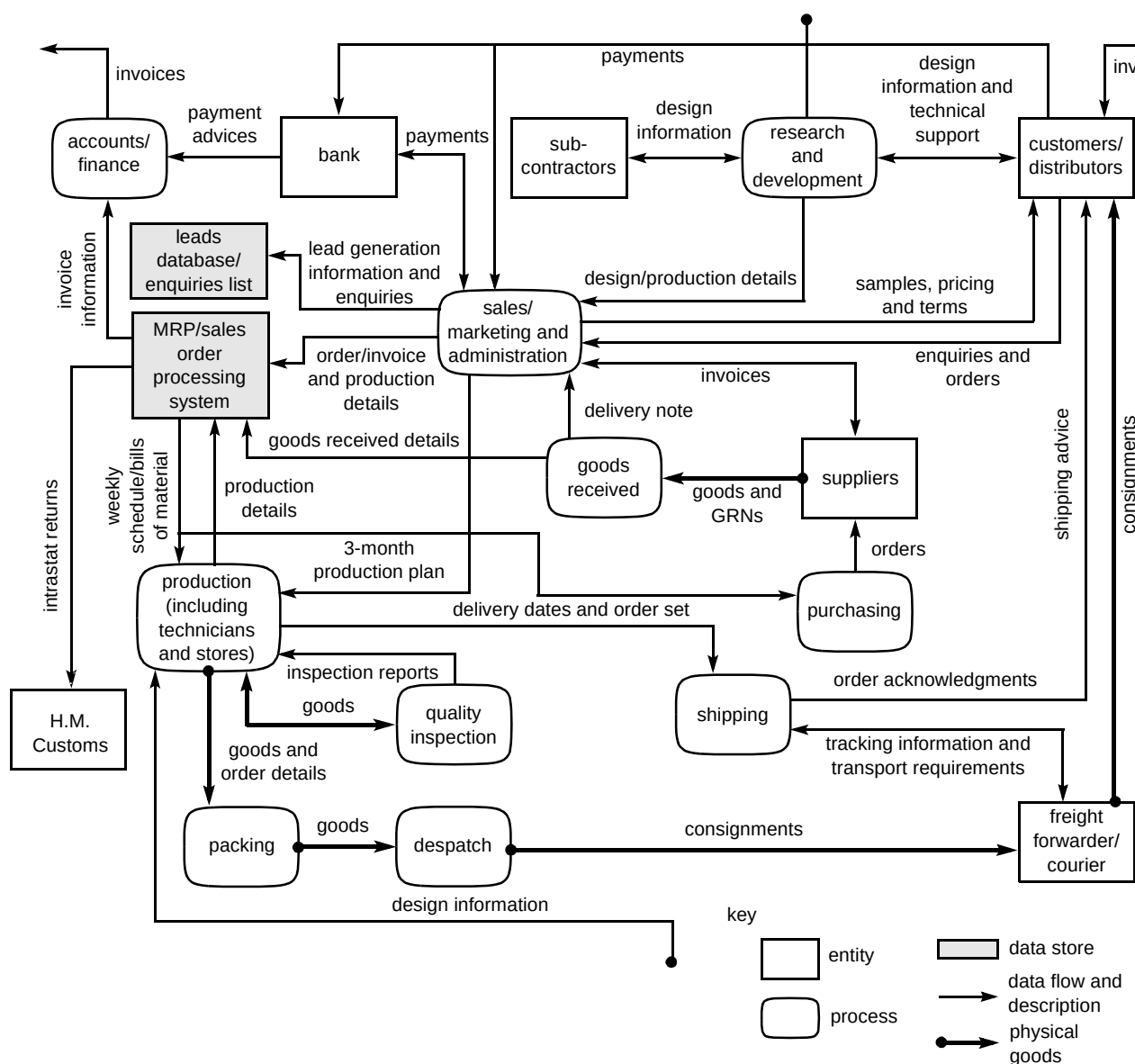


Fig 4 Major inter- and intra-company information flows at Company B.

efficient acquisition of products and development of relationships with suppliers and customers. As such an ISDN line, or fast dial-up line via modem, has proved important in providing fast access to the Internet. E-mail is a vital component in providing more efficient communication throughout the supply chain. These access arrangements on to the Internet have also allowed for an easier method of providing information about the company, more effective data exchanges, reduced administrative time and expenses, and the ability for researchers to access more quickly the information required during product development.

Indeed, for further confirmation the potential of the Web was recently recognised by Khoo, Tor and Lee [10] as '... rapidly becoming a powerful business tool because of its on-line commercial services and electronic commerce capabilities. The Web is poised to become the medium by which companies buy, trade, make contacts, exchange data

and information, discuss designs, and locate companies.' Attwood [11] highlights the Internet as now being established for commercial communications and forecasts [the Internet] to feature increasingly in both the social and working lives of most people. Attwood adds, '... the promise of a global electronic market-place is a siren's song to main board directors, and they are increasingly issuing directives to implement electronic commerce pilot projects.'

There has also been much written about EDI, now recognised as a specific subset of eCommerce. Recently, Cox and Ghoneim [12], Ricks [13] and Fearon and Philip [14] have emphasised the growth, and numerous associated benefits, of inter-organisation business document transmissions.

The benefits realised by three of the fifteen SMEs involved in the project are now highlighted.

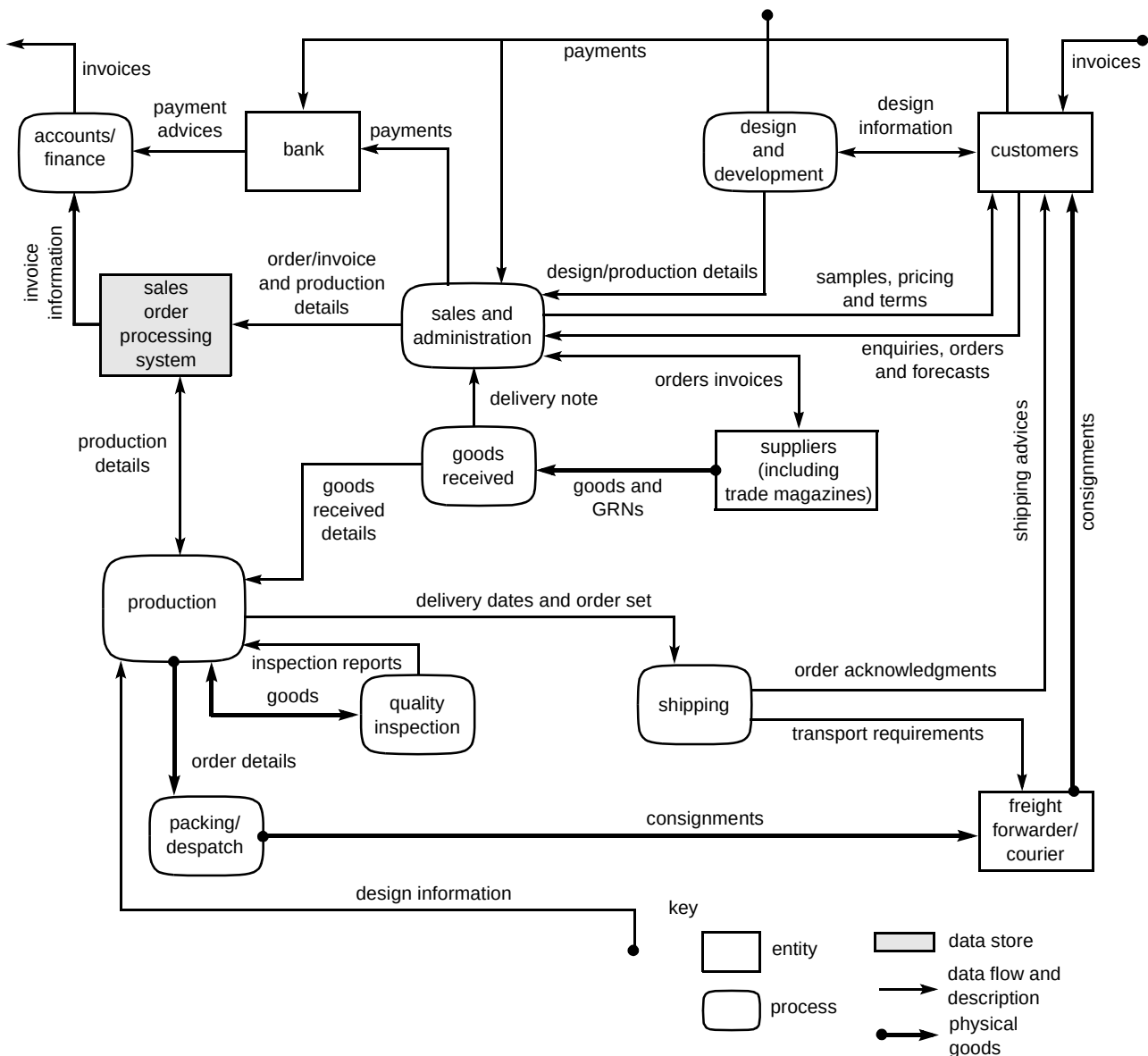


Fig 5 Major inter- and intra-company information flows at Company C.

4.1 Benefits for Company A

The marketing manager at Company A initiated a business process re-engineering (BPR) project in August 1997 that evaluated the entire company's operations by early 1998. The initial strategy falling out of this BPR exercise was a consolidation of the existing supplier base. This exercise has been largely influenced and facilitated by the Internet. The Internet has been used exclusively to source a team of consultants to assist with this project, as well as various new suppliers. The Internet has also been used to source non-standard supplies, such as air-conditioning and access platforms for computer rooms. In terms of benchmarking, Company A has examined multiple USA-based businesses (in their sector) for a sales and marketing strategy.

The activities of Far Eastern customers are now being tracked via the Pointcast service, which includes numerous

press releases and reports. The Internet-based Reuters information service is also being used specifically for Japanese and Korean customers (worth 40% of the company's turnover). This information is e-mailed to the marketing manager. In due course this will be extended to the entire customer base.

A good example of improved communications is where the company is now communicating with key suppliers (an estimated 40% of all suppliers and two customers are ordering PC-to-PC). Company A's customer base is high-tech advanced manufacturing and the marketing manager believes that being able to communicate with them electronically and understand the issues facing them has assisted in the generation of new business.

The marketing manager is increasingly using e-mail to develop relationships with numerous suppliers and customers, in addition to utilising the existing links with

credit agencies and for on-line banking. The company is now looking to implement eCommerce technologies to further enhance its position as a single source supplier, including the potential use of electronic catalogues and the development of extranets with major clients. Although no significant tangible benefits have been accrued to date, the marketing manager describes the Internet as '... a very cheap on-line consultant.'

4.2 Benefits for Company B

Company B increasingly uses the Internet for obtaining software upgrades, data sheets, technical information, competitor information and patent searches. The company recently provided a manual and software upgrade for its flagship ISDN testing product on its Web site which it asked a major customer's engineers to visit, and download from, if they wanted to complete the upgrade.

Traditionally the company would have been encouraged by the customer to provide its 1200 engineers with a hard copy. Each engineer's paper manual and software would have cost the company £6 to print and distribute, therefore costing £7200, as shown in Table 3.

However, Company B distributed just twelve copies of the upgrade pack to the customer's regional offices, with the remainder being downloaded from the Web. The cost of translating the manual into the hypertext mark-up language (HTML) and uploading the files on to the Web cost one engineer one hour of his time plus the twelve packs which were distributed. The cost of setting up a Web presence was £1811 for the first year. (A one-off £600 charge for the Web designer plus £130 for BT Internet access which includes the Web space. In addition, the ISDN installation, terminal adapter and rental charges have been included, a total of £1 081 for the first year).

This one-off solution therefore reflects a saving of £5305 and covers all of the costs incurred in setting up a Web presence and gaining Internet access, in addition to

providing invaluable customer service. Such uses of eCommerce technologies are now a core part of the company's strategy in terms of research and customer service.

By Spring 1998, Company B had therefore realised significant benefits from eCommerce. This led the company towards appointing a new employee to directly take responsibility for existing and future eCommerce activities. During the subsequent short period of employment, the company has received numerous e-mail enquiries, which significantly have generated multiple orders. The approximate value of these orders is £12 000, of which many are from overseas, as shown in Fig 6.

The payback period of the eCommerce investment was less than two months, as shown in Fig 7. Total profit during the eight months reached over £15 000.

4.3 Benefits for Company C

Company C now uses the Internet to source supplier and obtain information on competitors. The management team is increasingly using e-mail to enhance relationships with customers and suppliers. The company is now looking to implement eCommerce technologies to further develop existing relationships with customers and ultimately generate increased sales. The Web is also seen as a useful tool for providing information to customers for both development work and in order to keep customers updated on the company and its products.

For Company C, one major benefit realised from utilising the Internet appertains to a daily sales forecast/manufacturing plan from a major customer. This data exchange is received and processed in about thirty minutes with minimal manual intervention. The logistics manager estimates this electronic link, and internal automation, saves the company the equivalent of two days of a clerk's time per week, equating to £4600 per annum.

Table 3 Web site benefits for Company B.

Cost without eCommerce	£/annum	Cost with eCommerce	£/annum
Fixed costs: 1200 paper manuals, printing and distribution (@ £6/copy)*	7200	Fixed costs: 12 paper manuals, printing and distribution (@ £6/copy) ISDN line installation ISDN terminal adapter Rental BT Internet account Web site development/design	72 400 329 352 130 600
		Variable costs: Management/Engineer's time**	12
Total costs	7200	Total costs	1895
		Saving with eCommerce	5305

Notes:

* up-front design and development costs are sunk

** based on a £25 000 annual salary [15] and a 40-hour working week.

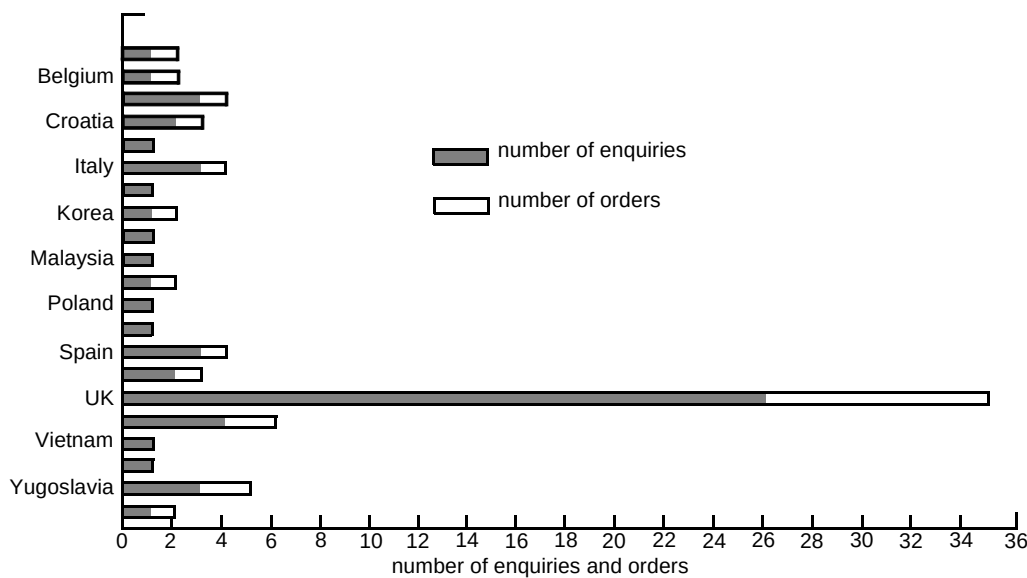


Fig 6 Enquiries and orders received by e-mail, March-July 1998.

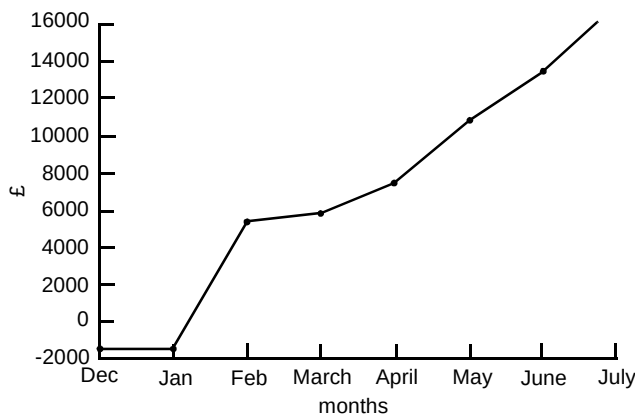


Fig 7 Company B's payback period from eCommerce technologies.

The cost to the company is a 9-minute phone call to their Internet service provider (ISP), CompuServe, each day. The saving to the company is therefore £2653 in the

first year following ISDN installation, as shown in Table 4, increasing to £3382 in each subsequent year (the £400 ISDN installation charge and terminal adapter are inapplicable after the first year).

However, there have also been three further fundamental benefits of this electronic link. Firstly, the administrative clerk who was mainly responsible for this activity, pre-eCommerce, has left the firm. This member of staff has not been directly replaced, but effectively the existing staff are covering the duties, such is the decrease in workload since the link has been in place. Company C are adding back a member of staff though, to the productive 'shop-floor' side of the business, therefore generating more production and at a lower staff cost (exact monetary values cannot be divulged).

Secondly, there has been a significant benefit to the customer in that what was a £400 000 stock-holding of Company C's 'looms' has now been reduced to just over

Table 4 Data exchange benefits for Company C.

Cost without eCommerce	£/annum	Cost with eCommerce	£/annum
Fixed costs: Two-fifths of a clerk's time {@£11 500 salary [16]}	4600	Fixed costs: ISDN line installation rental ISDN terminal adapter Clerk's time (1/2 an hour/day @ £5.53/hour*) CompuServe account	400 352 329 719 72
		Variable costs: ISDN call charges**	75
Total costs	4600	Total costs	1947
		Saving with eCommerce	2653

Notes:

* based on a 40-hour working week.

** based on a BT low-call charge rate of £0.10 for 3 minutes multiplied by 250 working days.

£200 000. The Logistics Manager believes this stock holding will continue to reduce, as it has already halved in the first nine months. The customer has also been able to clear nearly their entire backlog of Company C-produced stock.

Thirdly, this customer has now given 100% of its 'loom' orders to Company C which has meant an overall increase of about 10% turnover (again, the commercial, and financial, implications of this increase remain confidential).

This electronic link, and streamlining of internal processing, was initiated in March 1997 somewhat tentatively, according to the Logistics Manager. However, by May the benefits were becoming apparent and by August last year there were visible differences at the customer's site, where stocks were rapidly reducing. The essence of this solution has been the increase in the speed of communications, facilitated by a simple Internet e-mail link.

The payback period of this eCommerce investment, represented in Fig 8, was just less than three months. Following the initial £729 ISDN related charges, the significantly cheaper clerical input to the process meant that within three months savings were being made.

4.4 Summary of benefits

The growing importance of the various eCommerce technologies has been highlighted in three of the SMEs. All of the companies have accrued benefits to varying degrees. It is important to note that the impact of the technologies has affected not just one or two specific areas of the SMEs, but in fact the broad spectrum of pre-contract to post-contract activities. Furthermore, the role of the trading partners of each SME, throughout the supply chain, is key in realising these benefits. Table 5 highlights a number of benefits of eCommerce, as experienced by the SMEs, in their everyday activities with trading partners.

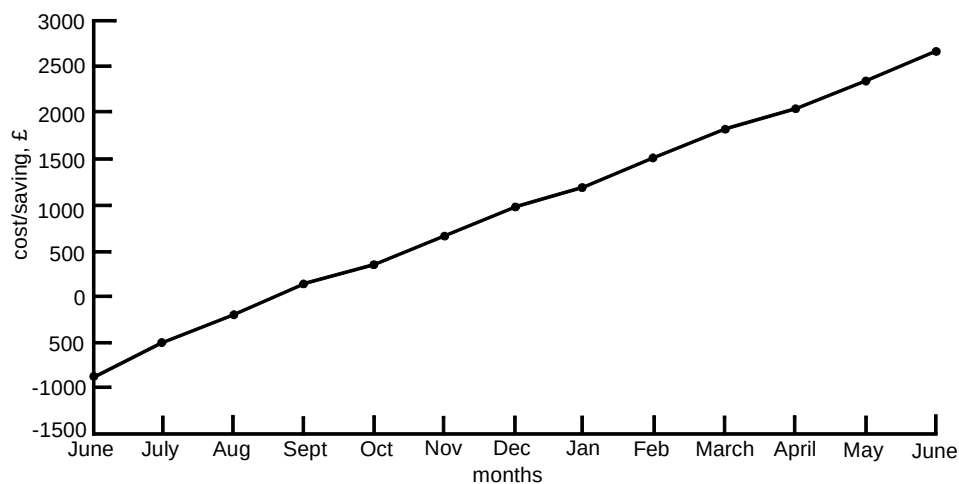


Fig 8 Company C's payback period for data exchanges.

Table 5 Benefits of eCommerce for SMEs in the supply chain.

Benefit	Electronic commerce technologies				
	Web site/access	EDI	E-catalogue	E-banking	Desktop VC
Increased research capabilities (e.g. Company B)	✓				✓
More efficient sourcing/purchasing (e.g. Company A)	✓				
Reduced administration time/expense (e.g. Company C)	✓	✓	✓	✓	
More effective data exchanges (e.g. Company C)		✓			✓
Increased visibility and sales (e.g. Company B)	✓				
Customer service tool (e.g. Company B)	✓		✓		✓

The most important project finding is that all of the 15 SMEs have found some benefit in the eCommerce technologies implemented. Whereas all parties involved would of course have preferred to see a proliferation of tangible benefits during the first year of the project, the reality has been a slower take-up and development of ideas for which the SMEs themselves can use the technologies. However, there have been numerous quasi-tangible and intangible benefits accrued. It is these benefits that are the most significant finding.

The implications of eCommerce on the SMEs have been fundamental. In the opening paragraphs of this paper it was stated that the Internet has become core to a number of the activities in all of the SMEs involved in the project. As has been highlighted in this section, this is no exaggeration. All of the SMEs now regularly use e-mail as a communications tool, increasingly superseding that of traditional telephone and fax reliance. The other applications highlighted also exemplify how eCommerce is enabling these SMEs to gain advantages throughout the supply chain.

4.5 *Applicability to the population*

The case studies clearly provide only a snapshot of a very small sample of SMEs. But the question remains how representative of the population is such a small sample of companies, both in the UK and beyond. The answer lies in the recruitment process of SMEs for this project. Recruitment of the fifteen SMEs was undertaken over the course of a nine-month period in which time-specific criteria were used.

In addition to covering the geographic catchment area of ISW, the SMEs were selected to provide a range of:

- domestic and international traders,
- business-to-business and business-to-consumer traders,
- service providers and product manufacturers,
- at least one cluster of companies within the fifteen which trade together, i.e. intra-business.

The SMEs cover a number of different industry types, with thirteen manufacturers — chemicals, electrical and electronic engineering, metals, footwear/clothing, rubber/plastics, giftware and other, and the final two in industrial supplies and transport/communication.

Most of the SMEs are privately owned — typical for SMEs — twelve are limited companies with the exception of one privately-held and two which are owned by public limited companies. The turnover range is from just £100 000 to some £20m and the employee range is from 4 to 170, the smaller in both cases being the same firm. The aggregated turnover is £49.1m, therefore averaging at

£3.3m, and the aggregated employee count is 769, an average of 51. Thus the SMEs were recruited to provide a representative cross-section of the South Wales industrial base, and the wider UK economy, according to other internal (non-published) research undertaken by eCIC.

The authors therefore believe that the findings of this research project can be generalised to the wider, global SME population. There was very little bias in the recruitment process other than the SMEs themselves deciding to become involved in the project. The most successful applications of eCommerce have been realised where the SMEs in question have displayed a positive attitude towards the technologies, have been willing to discuss their business role within the supply chain, and have been willing to experiment.

5. **Problems of electronic commerce for SMEs**

A variety of different issues and everyday problems have been highlighted. Figure 9 shows the result of a survey conducted by the authors — the purpose being to highlight specific issues and to ask the SMEs whether they agreed such issues were pertinent for their firm. The issues themselves were compiled following the early discussions with the SMEs. This survey was conducted via a combination of e-mail and telephone conversations with each SME during late 1997.

It is interesting to note that across the trading model, depicted in Fig 10, the spread of issues identified was fairly even. However, the monetary cost and savings element of each (i.e. weighting) may well prove uneven once calculated at the end of the project. Early indications, as already highlighted, are that the largest issues in terms of financial and human cost (or potential cost) are those in the pre- and post-contract stages.

Small businesses do not have the resources of medium-sized to large companies, or corporations, with which they may trade every day. This factor hinders the entire business of an SME in that the required, or preferred, investment in skills and technologies cannot be met. This is exemplified where the constant issue of desirability against disability has been raised. Figure 9 highlights the higher response levels of problems encountered in pre-eCommerce trading among the SMEs.

A 1997 KPMG management consulting survey [17] identified technological concerns as no longer being the uppermost barriers to implementing eCommerce. In a comparative study during 1996 Baker [17] found 'cost' and 'security' to be ranked second and third among spontaneously mentioned barriers (17% and 13% respectively) after 'resistance' (21%).

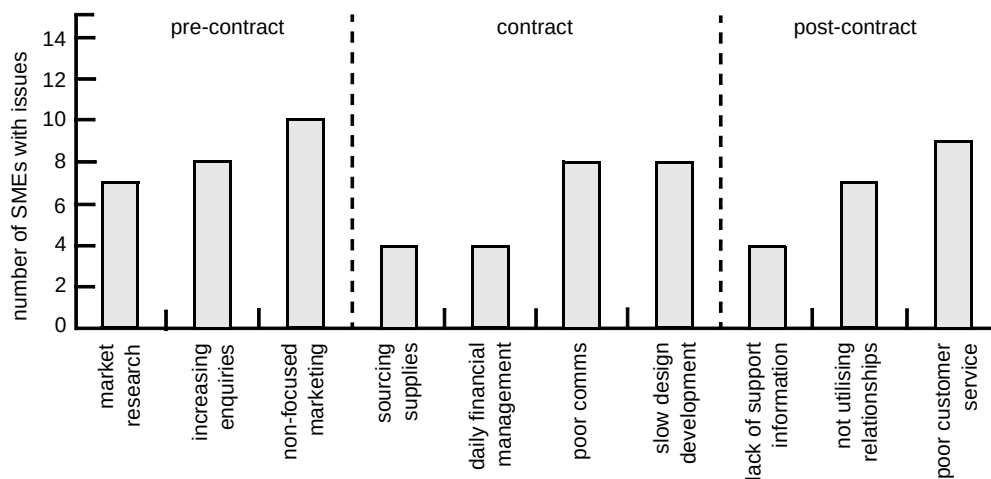


Fig 9 Number of problems encountered by each SME.

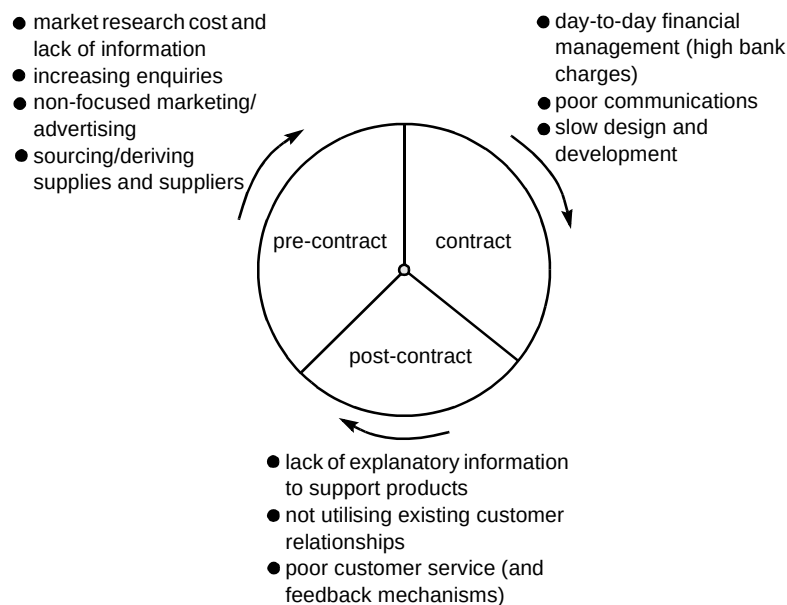


Fig 10 Common issues of pre-eCommerce trading among SMEs.

This survey mirrors the issues arising from the eCommerce test platform in South Wales. It is the resistance to trying new technologies that has been the biggest hurdle, as depicted in Fig 11. The fact that that SMEs are uncertain of technology and fearful of cost, is poignant, as noted by Lowell Mooney and Pittman [18]. The requirement for a critical mass of potentially connected customers is also necessary before eCommerce becomes a reality.

Pollard and Hayne [19] provide a more, detailed account of the wider information systems issues that SMEs face. The authors argue that the key issues facing small businesses have changed over time and are different from those facing large businesses.

However, on a more positive note, recognition should be given to the UK government ‘... which is trying to use multiple methods available to pass on the message of eCommerce benefits to SMEs’ [2].

6. eCommerce requirements of SMEs and integration into the supply chain

Feedback from the SMEs has provided a number of requirements for eCommerce. For eCommerce to become widespread and applicable to the needs of SMEs and their operations within the supply chain, the following is almost an aggregate ‘wish-list’:

- to find new suppliers and therefore sources for designs and products,
- to obtain time savings on present operations especially in visits to companies,
- to increase customer contact and gain better customer perception,
- to increase the level of competence in eCommerce with more staff able to use these technologies.

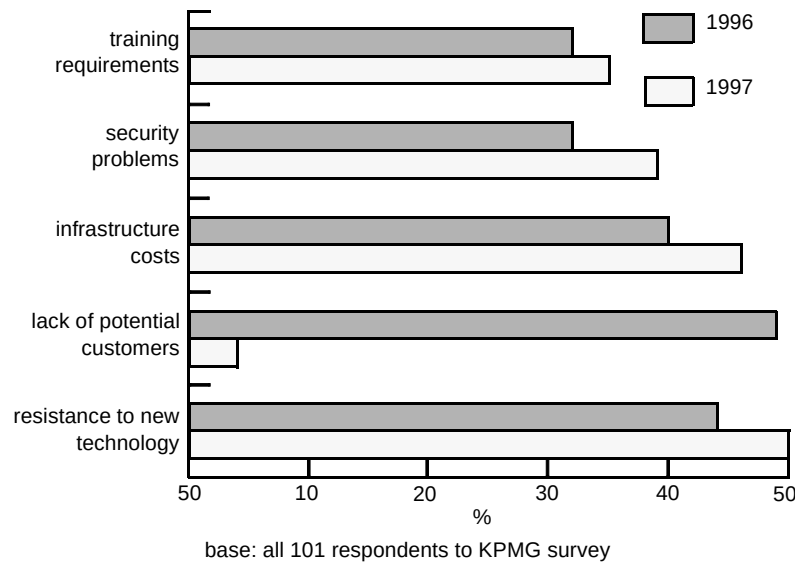


Fig 11 Most significant barriers to eCommerce.

These requirements necessitate two vital levels of support from industry and government. Firstly, it is imperative that the private sector understands the generic needs of SMEs and attempts to provide products and services, which closely match these. Secondly, education and training are vital to ensuring that SMEs understand the role eCommerce can play in enhancing their businesses and ultimately integrating them more deeply into the supply chain.

This requirement for supporting the development and take-up of technologies by SMEs is largely understood. For example, among many others Morgan and Crawford [20], Lauder and Westall [21], Brouchet [22], Hyvärinen [23], Bannock and Daly [24], the Organisation for Economic Co-operation and Development [25, 26], Irving [27], Goldsworthy [28] and Poon and Swatman [29] have all advocated this requirement. The UK government has of course also included SMEs high on its Information Society Initiative [30].

With regards to developing software for manufacturing systems, Little [31] suggests that SMEs ‘... rarely have the IT infrastructure and experience to exploit more formal structured methods,’ let alone for eCommerce applications. Finally, Levy and Powell [32] discuss the flexibility of SMEs *vis-à-vis* information systems.

Perhaps the most recognisable role of SMEs in a supply chain, however, is that of suppliers. All of the SMEs involved in the project have a small number of major clients generating a large proportion of revenues. In addition, most have a much larger number of customers providing a smaller percentage of revenues. This is typical of SMEs, as identified in a study by the venture capital firm, 3i [33], in which 70% of the small firms interviewed obtained over 50% of their revenues from their top two customers. By

integrating them into the supply chain, eCommerce technologies can offer significant benefits.

From the days of EDI becoming a term of trade in a number of sectors [34], the wider remit of eCommerce now offers SMEs the ability to build and develop relationships with key customers and to ultimately offer better service at all stages of the trading model.

The vital component of eCommerce in this integration is access to, and utilisation of, the Internet. However, SMEs need to look inwardly at their own requirements before simply implementing technology for technology’s sake. The most obvious areas of potential benefit should be the primary focus. From the test platform these areas are at the stages of pre-contract (i.e. research and design, negotiation) and post-contract (i.e. customer service).

7. Conclusions

This paper has shown how SMEs can actively use eCommerce technologies in order to cement, and develop, their roles in the supply chain. Three specific case studies have highlighted how eCommerce, and specifically the Internet, have become core to the activities of the companies.

The major benefits that eCommerce can provide to SMEs in their interactions with trading partners are faster communication, more effective dissemination and collection of information, and closer relationships throughout the supply chain. The major issues are technology incompatibilities and a lack of understanding of potential benefits.

The most important requirements of eCommerce to SMEs are access to information about eCommerce and generic products/services, which can be easily implemented.

Acknowledgements

This research utilised reports, studies, surveys from the Organisation for Economic Co-operation and Development [24, 25, 35] the Cardiff University Business School and Western Mail newspaper [36, 37], the Welsh Development Agency [38–40], the Welsh Chamber of Commerce [41] and other authors (e.g. Bannock and Daly [24]). The authors also acknowledge the support of BT during this research project.

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